# Logic!

COMP0147

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# Logic: Introduction

- **Logic** is the branch of philosophy, mathematics, and computer science that deals with **valid inferences**.

  Logic is the foundation of mathematical reasoning and all automated reasoning.

- It studies the **preservation of truth** during an argument.

  Logic concerns techniques that ensure that:

  1. starting from true hypotheses,
  2. we always reach conclusions that are also true.

- The rules of logic give precise meaning to mathematical statements.

  They are essential in the construction of **mathematical proofs**.

# Logic: Introduction

- In addition to its importance in mathematics, logic has applications in computer science in areas such as:

  1. circuit design,
  2. computer programming,
  3. artificial intelligence,
  4. automatic theorem proving,
  5. program correctness verification,
  6. ...

- In this class we will see:
  - propositional logic,
  - first-order logic,
  - (some) proof theory,
  - (some) proof techniques,
  - and fallacies as a bonus track!

# Propositional Logic

# Propositions

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  - "$1 + 1 = 2$."
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- The following sentences are not propositions:
  - "What time is it?"
  - "Study hard for the exam."
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- The following sentences are not propositions:
  - "What time is it?" (Not a declarative sentence.)
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  - "$x + 2 = 3$."

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# Propositions

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- The following sentences are not propositions:
  - "What time is it?" (Not a declarative sentence.)
  - "Study hard for the exam." (Not a declarative sentence.)
  - "$x + 2 = 3$." (Neither true nor false.)

# Propositions

- We use letters to denote **propositional variables**, that is, variables that represent propositions:

$$p,\ q,\ r,\ s,\ t,\ \ldots$$

- The **truth value** of a proposition can be:
  - **true**, denoted by $V$ (true) or $T$ (true), or
  - **false**, denoted by $F$ (false or false).
- The area of logic that deals with propositions is called **propositional calculus** or **propositional logic**.

# Compound Propositions

- **Compound propositions** can be created by combining existing propositions.

  The combination of propositions is done using **logical operators** or **logical connectives** such as:

  - negation (not),
  - conjunction (and),
  - disjunction (or),
  - implication (implies),
  - biconditional (if and only if).

- We will now study these connectives.

# Propositions: Negation

- Let $p$ be a proposition.

  The **negation of** $p$, denoted by $\neg p$ (or also $\overline{p}$), is the statement

  "It is not the case that $p$."

  The proposition $\neg p$ is read as "not $p$".

  The truth value of $\neg p$ is the opposite of the truth value of $p$.

- **Truth table** for the negation of a proposition $p$:

**Negation**

| $\mathbf{p}$ | $\neg\mathbf{p}$ |
|---|---|
| $T$ | $F$ |
| $F$ | $T$ |

# Logical Connectives: Conjunction

- Let $p$ and $q$ be propositions.

  The **conjunction of $p$ and $q$**, denoted by $p \wedge q$, is the statement

  "$p$ and $q$".

  The conjunction $p \wedge q$ is true when both $p$ and $q$ are true, and false otherwise.

- **Truth table** for the conjunction of two propositions $p$ and $q$:

**Conjunction**

| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{p \wedge q}$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ |
| $F$ | $F$ | $F$ |

# Logical Connectives: Disjunction

- Let $p$ and $q$ be propositions.

  The **disjunction of $p$ and $q$**, denoted by $p \vee q$, is the statement

  "$p$ or $q$".

  The disjunction $p \vee q$ is true when at least one of $p$ or $q$ is true, and false otherwise.

- **Truth table** for the disjunction of two propositions $p$ and $q$:

**Disjunction**

| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{p \vee q}$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $T$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $F$ |

# Logical Connectives: Conditional Propositions

- **Truth table** for the conditional proposition involving two propositions $p$ and $q$:

**Implication**

| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{p \rightarrow q}$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $T$ |
| $F$ | $F$ | $T$ |

The conditional statement $p \rightarrow q$ is false when $p$ is true and $q$ is false, and true otherwise.

# Conditional Propositions in Natural Language

- Implications appear in mathematics and natural language in several forms.

  The conditional statement $p \rightarrow q$ can be expressed as:

  - "if $p$, then $q$"
  - "if $p$, $q$"
  - "$q$ if $p$"
  - "$q$ when $p$"
  - "$p$ is sufficient for $q$"
  - "$q$ is necessary for $p$"
  - "$q$ unless $\neg p$"
  - "$p$ implies $q$"
  - "$p$ only if $q$"
  - "a sufficient condition for $q$ is $p$"
  - "a necessary condition for $p$ is $q$"
  - "$q$ whenever $p$"
  - "$q$ follows from $p$"

# Conditional Propositions in Natural Language

- Let the propositions be:

$$p : \text{“It is sunny.”}$$

$$q : \text{“I will go to the club.”}$$

  The implication $p \rightarrow q$ can be written in natural language as:

  - “If it is sunny, I will go to the club.”
  - “Being sunny is a sufficient condition for me to go to the club.”
  - “I will go to the club unless it is not sunny.”
  - “The fact that I go to the club follows from the fact that it is sunny.”
  - “I will go to the club whenever it is sunny.”
  - “It is sunny only if I go to the club.”

# Conditional Propositions: Converse, Contrapositive, and Inverse

- Given an implication $p \to q$:
  - its **converse** is the implication $q \to p$,
  - its **contrapositive** is the implication $\neg q \to \neg p$,
  - its **inverse** is the implication $\neg p \to \neg q$.

# Conditional Propositions: Converse, Contrapositive, and Inverse

- Let the proposition be:

  "The home team wins whenever it is raining."

  This implication can be written as $p \rightarrow q$, where

  $p$ is the proposition "It is raining", and

  $q$ is the proposition "The home team wins".

  - The converse $q \rightarrow p$ is the proposition

    "If the home team wins, it is raining."

  - The contrapositive $\neg q \rightarrow \neg p$ is the proposition

    "If the home team does not win, then it is not raining."

  - The inverse $\neg p \rightarrow \neg q$ is the proposition

    "If it is not raining, the home team does not win."

# Logical Connectives: Biconditional Propositions

- Let $p$ and $q$ be propositions.

  The **biconditional statement** or **double implication** $p \leftrightarrow q$ is the statement

  "$p$ if and only if $q$".

  The biconditional statement $p \leftrightarrow q$ is true when $p$ and $q$ have the same truth value, and false otherwise.

- In natural language, $p \leftrightarrow q$ is often expressed as:
  - "$p$ is necessary and sufficient for $q$."
  - "$p$ iff $q$." Note that we use "iff" with two "f"s.

# Logical Connectives: Biconditional Propositions

- **Truth table** for the biconditional of two propositions $p$ and $q$:

**Double Implication**

| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{p \leftrightarrow q}$ |
|---|---|---|
| $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ |
| $F$ | $F$ | $T$ |

- The biconditional proposition $p \leftrightarrow q$ is true whenever both $p \rightarrow q$ and $q \rightarrow p$ are true, and false otherwise.

# Truth Table for Compound Propositions

- We have introduced negation and the logical connectives of disjunction, conjunction, exclusive or, implication, and biconditional.
- We can use these operators to express increasingly complex propositions.
- To determine the truth value of compound propositions, we can use **truth tables**.

Truth table for the expression $(p \vee \neg q) \rightarrow (p \wedge q)$:

| $\mathbf{p}$ | $\mathbf{q}$ | $\neg\mathbf{q}$ | $\mathbf{p} \vee \neg\mathbf{q}$ | $\mathbf{p} \wedge \mathbf{q}$ | $(\mathbf{p} \vee \neg\mathbf{q}) \rightarrow (\mathbf{p} \wedge \mathbf{q})$ |
|---|---|---|---|---|---|
| $T$ | $T$ | $F$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $T$ | $T$ | $F$ | $F$ |
| $F$ | $T$ | $F$ | $F$ | $F$ | $T$ |
| $F$ | $F$ | $T$ | $T$ | $F$ | $F$ |

# Order of Precedence of Logical Operators

- In a compound expression, the order of operator application is:
  1) negation: $\neg$
  2) conjunction: $\wedge$
  3) disjunction: $\vee$
  4) implication: $\rightarrow$
  5) biconditional: $\leftrightarrow$
- Examples:
  1. $p \vee \neg q \wedge r$ is equivalent to $p \vee ((\neg q) \wedge r)$.
  2. $p \rightarrow q \vee r$ is equivalent to $p \rightarrow (q \vee r)$.

# Bitwise Logical Operators

- Computers represent information using **bits**, which are binary digits 0 or 1.

  The bit 1 represents $T$ (true), and the bit 0 represents $F$ (false).

- Logical connectives can be used to operate on **Boolean variables**, whose values are bits.

  Conjunction is called **AND**, disjunction is called **OR**, and exclusive or is called **XOR** (*exclusive or*).

  Truth table for the operators AND, OR, and XOR:

| x | y | x **AND** y | x **OR** y | x **XOR** y |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 |

# Bitwise Logical Operators

- A **bit string** is a sequence of zero or more bits.

  The length of a string is the number of bits in the string.

- Examples:

  1. 101 is a bit string of length 3.
  2. 0100111 is a bit string of length 7.

- Binary operators can be applied **bit by bit** to two bit strings of the same length. The result is a string of the same length.

-

| | |
|---|---|
| | 0110 |
| | 1101 |
| result of bitwise AND: | 0100 |
| result of bitwise OR: | 1111 |
| result of bitwise XOR: | 1011 |

# Logical Equivalences

- Show that $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.

  **Solution.** Truth table for $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$:

| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{r}$ | $\mathbf{q \wedge r}$ | $\mathbf{p \vee (q \wedge r)}$ | $\mathbf{p \vee q}$ | $\mathbf{p \vee r}$ | $\mathbf{(p \vee q) \wedge (p \vee r)}$ |
|---|---|---|---|---|---|---|---|
| $T$ | $T$ | $T$ | $T$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $T$ | $F$ | $F$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $T$ | $F$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $F$ | $T$ | $T$ | $T$ | $T$ |
| $F$ | $T$ | $T$ | $T$ | $T$ | $T$ | $T$ | $T$ |
| $F$ | $T$ | $F$ | $F$ | $F$ | $T$ | $F$ | $F$ |
| $F$ | $F$ | $T$ | $F$ | $F$ | $F$ | $T$ | $F$ |
| $F$ | $F$ | $F$ | $F$ | $F$ | $F$ | $F$ | $F$ |

  Since the columns corresponding to $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ always have the same truth value, $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

  The equivalence we demonstrated is the **law of distribution of disjunction over conjunction**.

# Some important logical equivalences

| Name | Equivalence |
|---|---|
| Identity Laws | $p \wedge T \equiv p$<br>$p \vee F \equiv p$ |
| Domination Laws | $p \wedge F \equiv F$<br>$p \vee T \equiv T$ |
| Idempotent Laws | $p \wedge p \equiv p$<br>$p \vee p \equiv p$ |
| Double Negation Law | $\neg(\neg p) \equiv p$ |
| Commutative Laws | $p \wedge q \equiv q \wedge p$<br>$p \vee q \equiv q \vee p$ |
| Associative Laws | $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$<br>$(p \vee q) \vee r \equiv p \vee (q \vee r)$ |
| Distributive Laws | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$<br>$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ |
| De Morgan's Laws | $\neg(p \wedge q) \equiv \neg p \vee \neg q$<br>$\neg(p \vee q) \equiv \neg p \wedge \neg q$ |
| Absorption Laws | $p \vee (p \wedge q) \equiv p$<br>$p \wedge (p \vee q) \equiv p$ |
| Negation Laws | $p \wedge \neg p \equiv F$<br>$p \vee \neg p \equiv T$ |

# Logical Equivalences

- Logical equivalences involving conditional propositions:

| **Equivalences** |
|---|
| $p \to q \equiv \neg p \vee q$ |
| $p \to q \equiv \neg q \to \neg p$ |
| $p \vee q \equiv \neg p \to q$ |
| $p \wedge q \equiv \neg(p \to \neg q)$ |
| $\neg(p \to q) \equiv p \wedge \neg q$ |
| $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$ |
| $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$ |
| $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$ |
| $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$ |

- Logical equivalences involving biconditional propositions:

| **Equivalences** |
|---|
| $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$ |
| $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ |
| $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ |
| $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$ |

# Uses of De Morgan's Laws

- The two logical equivalences known as De Morgan's Laws are particularly important.
- The law

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

states that the negation of a disjunction is the conjunction of the negations.

- Use De Morgan's Laws to negate the proposition:

"Julia goes to the cinema or Catarina goes to the cinema."

**Solution.** This proposition can be written as $p \vee q$, where $p$ is "Julia goes to the cinema" and $q$ is "Catarina goes to the cinema".

By De Morgan's Law, the negation is $\neg(p \vee q) \equiv \neg p \wedge \neg q$, which translates to:

"Julia does not go to the cinema and Catarina does not go to the cinema."

# Uses of De Morgan's Laws

- De Morgan's Law

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

states that the negation of a conjunction is the disjunction of the negations.

- Use De Morgan's Laws to negate the proposition:

"Pedro ate lasagna and cake."

**Solution.** This proposition can be written as $p \wedge q$, where $p$ is "Pedro ate lasagna" and $q$ is "Pedro ate cake".

By De Morgan's Laws, the negation is $\neg(p \wedge q) \equiv \neg p \vee \neg q$, which translates to:

"Pedro did not eat lasagna or he did not eat cake."

# Building New Logical Equivalences

- As we have seen, truth tables can be used to verify logical equivalences.
- In general, for expressions involving $n$ logical variables, $2^n$ lines are needed in the table.

  For large $n$, it may be inconvenient to build the truth table.
- An alternative is to use already known logical equivalences to derive new ones directly.

# Building New Logical Equivalences

- Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

  **Solution.**

  $$\neg(p \rightarrow q) \equiv$$

# Building New Logical Equivalences

- Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent.

  **Solution.**

  $$\neg(p \to q) \equiv \neg(\neg p \vee q) \qquad \text{(from the table of conditional equivalences)}$$

# Building New Logical Equivalences

- Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$\begin{aligned} \neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{(from the table of conditional equivalences)} \\ &\equiv \neg(\neg p) \wedge \neg q && \text{(by De Morgan's Laws)} \end{aligned}$$

# Building New Logical Equivalences

- Show that $\neg(p \to q)$ and $p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$\begin{aligned}
\neg(p \to q) &\equiv \neg(\neg p \vee q) && \text{(from the table of conditional equivalences)} \\
&\equiv \neg(\neg p) \wedge \neg q && \text{(by De Morgan's Laws)} \\
&\equiv p \wedge \neg q && \text{(by the double negation law)}
\end{aligned}$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

  $$\neg(p \vee (\neg p \wedge q)) \equiv$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

  $$\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)) \qquad \text{(by De Morgan's Laws)}$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$\begin{aligned} \neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(by De Morgan's Laws)} \\ &\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \end{aligned}$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(by the double negation law)}
\end{aligned}
$$

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&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(by the double negation law)} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{(by the distributive law)}
\end{aligned}
$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(by the double negation law)} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{(by the distributive law)} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{(since } \neg p \wedge p \equiv F\text{)}
\end{aligned}
$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(by the double negation law)} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{(by the distributive law)} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{(since } \neg p \wedge p \equiv F\text{)} \\
&\equiv (\neg p \wedge \neg q) \vee F && \text{(by the commutative law)}
\end{aligned}
$$

# Building New Logical Equivalences

- Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

  **Solution.**

$$
\begin{aligned}
\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (\neg(\neg p) \vee \neg q) && \text{(by De Morgan's Laws)} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{(by the double negation law)} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{(by the distributive law)} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{(since } \neg p \wedge p \equiv F\text{)} \\
&\equiv (\neg p \wedge \neg q) \vee F && \text{(by the commutative law)} \\
&\equiv \neg p \wedge \neg q && \text{(by the identity law)}
\end{aligned}
$$

# Building New Logical Equivalences

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  **Solution.**

  $$(p \wedge q) \rightarrow (p \vee q) \equiv$$

# Building New Logical Equivalences

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  **Solution.**

  $$(p \wedge q) \rightarrow (p \vee q) \equiv \neg(p \wedge q) \vee (p \vee q) \qquad \text{(conditional equivalence)}$$

# Building New Logical Equivalences

- Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

  **Solution.**

$$
\begin{aligned}
(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) && \text{(conditional equivalence)} \\
&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{(by De Morgan's Laws)}
\end{aligned}
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# Building New Logical Equivalences

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&\equiv (\neg p \vee \neg q) \vee (p \vee q) && \text{(by De Morgan's Laws)} \\
&\equiv (\neg p \vee p) \vee (\neg q \vee q) && \text{(commutativity and associativity)} \\
&\equiv T \vee T && \text{(by the negation law)} \\
&\equiv T && \text{(by the domination law)}
\end{aligned}
$$

# Predicate Logic: Introduction

- Propositional logic, which we have studied so far, covers the analysis of compound propositions, i.e., simple propositions connected by connectives $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, ...
- However, propositional logic is not expressive enough to capture all mathematical statements.
- For example, suppose we know that

  1 "All students enrolled in Discrete Mathematics are dedicated students", and that

  2 "Alice is enrolled in Discrete Mathematics."

  Intuitively, we can then conclude that

  3 "Alice is a dedicated student."

  This type of inference is only possible once we introduce the concept of predicates and quantifiers, which is what we will do here.

# First-order Logic (FOL)

# Predicates: Introduction

- Statements such as the following are common in mathematics:
  1. $x \geq 12$,
  2. $x + y = z$,
  3. The student $x$ got the highest grade in the class on the test.
- Such statements are written in terms of variables.

  Unless the values of the variables are specified, the statements are neither true nor false.

  As they have no truth value, they are not propositions.

# Predicates: Introduction

- The statement

  "$x$ is a real number"

  can be divided into two parts:

  1. the first part, the **variable** $x$, is the subject of the statement,
  2. the second part "is a real number" is a **predicate**, that is, a property that the subject of the statement may or may not satisfy.

  The statement "$x$ is a real number" can have different truth values depending on the value assigned to the variable $x$.

  - When $x = \pi$, the statement "$x$ is a real number" is true.
  - When $x = \sqrt{-2}$, the statement "$x$ is a real number" is false.

# Predicates: Introduction

- We can view this statement as a **propositional function**

  $P(x)$: "$x$ is a real number"

  which maps values of $x$ to truth values (true or false):

  - $P(\pi)$ is true.
  - $P(\sqrt{-2})$ is false.

# Predicates

- A **predicate** is a sentence that contains a finite number of variables and becomes a proposition when the variables are replaced by specific values.

  Intuitively, predicates assign qualities to subjects; they relate subjects to each other or relate subjects to objects.

- Predicates are classified according to the number of their variables.

  A predicate $P(x_1, x_2, \ldots, x_n)$ with $n$ variables is called an **n-ary predicate**.

# Predicates: Examples

- Unary (1-ary) predicate:

  $P(x)$: "$x$ is a prime number."

  When we assign $x = 5$, $P(5)$ is true.

  When we assign $x = 6$, $P(6)$ is false.

- Binary (2-ary) predicate:

  $Q(x, y)$: "$x$ is greater than $y$."

  $Q(5, 3)$ is true, but $Q(2, 4)$ is false.

- Ternary (3-ary) predicate:

  $R(x, y, z)$: "$x + y = z$."

  $R(2, 3, 5)$ is true, but $R(2, 3, 6)$ is false.

# Predicates and Propositions

- When specific values are assigned to all variables of a predicate, the resulting statement becomes a proposition.
- Example:

  $$P(x)\text{: } \text{“}x > 3\text{”}$$

  Assigning $x = 5$ gives the proposition $P(5)$, which is true.

  Assigning $x = 2$ gives the proposition $P(2)$, which is false.
- The set of all values for which $P(x)$ is true is called the **truth set** of the predicate $P(x)$.

$$P(x)\text{: } \text{“}x > 3\text{”}$$
Domain: integers
Truth set: $\{4, 5, 6, 7, ...\}$

# Domain of Discourse

- The set of possible values that a variable may assume is called the **domain of discourse** (or simply the **domain**) of that variable.
- The truth value of a predicate depends on the domain chosen.
- Example:

$$P(x)\text{: } \text{“}x > 9\text{”}$$

  - If the domain is the set of natural numbers, then $P(10)$ is true and $P(5)$ is false.
  - If the domain is $\{2, 4, 6, 8\}$, then $P(x)$ is false for all $x$ in the domain.

# Predicates: Summary

- A predicate $P(x_1, x_2, \ldots, x_n)$ is a statement involving variables $x_1, x_2, \ldots, x_n$ that becomes a proposition when specific values are assigned to the variables.
- The truth or falsity of $P(x_1, x_2, \ldots, x_n)$ depends on the values assigned to the variables and on the domain of discourse.
- The set of values that make the predicate true is called its **truth set**.

# Quantifiers: Introduction

- Predicates allow us to describe properties of elements in a domain.

  To make general statements about ÒallÓ or ÒsomeÓ elements of a domain, we use **quantifiers**.

- Quantifiers indicate the extent to which a predicate holds true over the domain of discourse.

- The two main quantifiers are:
  - the **universal quantifier**, and
  - the **existential quantifier**.

# Universal Quantifier

- The **universal quantifier**, denoted by $\forall$, expresses that a predicate holds for all elements of the domain.
- The statement

  $$\forall x\, P(x)$$

  is read as

  "For all $x$, $P(x)$ is true."
- $\forall x\, P(x)$ is true if $P(x)$ is true for every element $x$ in the domain of discourse, and false otherwise.

# Universal Quantifier: Example

- Let the domain be the set of natural numbers, and let

  $$P(x)\text{: } \text{“}x + 1 > x\text{”}.$$

  Then, the statement

  $$\forall x\, P(x)$$

  is true, since for every natural number $x$, $x + 1 > x$.

- If we take

  $$Q(x)\text{: } \text{“}x^2 < x\text{”}$$

  over the domain of positive integers, then $\forall x\, Q(x)$ is false, since, for example, $Q(2)$ is false.

# Existential Quantifier

- The **existential quantifier**, denoted by $\exists$, expresses that a predicate holds for at least one element of the domain.
- The statement

$$\exists x\, P(x)$$

is read as

"There exists an $x$ such that $P(x)$ is true."

- $\exists x\, P(x)$ is true if $P(x)$ is true for at least one element $x$ in the domain of discourse, and false otherwise.

# Existential Quantifier: Example

- Let the domain be the set of integers, and let

  $$P(x)\text{: } \text{“}x^2 = 4\text{”}.$$

  Then, the statement

  $$\exists x\, P(x)$$

  is true, since there are values of $x$ (namely $x = 2$ and $x = -2$) for which $x^2 = 4$.

- If we let

  $$Q(x)\text{: } \text{“}x^2 = -1\text{”}$$

  over the domain of real numbers, then $\exists x\, Q(x)$ is false, since there is no real number $x$ such that $x^2 = -1$.

# Quantifiers: Comparison

| Quantifier | Symbol | Meaning |
|---|---|---|
| Universal | $\forall$ | For all elements of the domain |
| Existential | $\exists$ | There exists at least one element in the domain |

- $\forall x\, P(x)$ means that $P(x)$ is true for every $x$ in the domain.
- $\exists x\, P(x)$ means that there is at least one $x$ in the domain for which $P(x)$ is true.

# Quantifiers: Examples

- Let the domain be the set of all real numbers.
  1. $P(x)$: "$x^2 \geq 0$"
     - $\forall x\, P(x)$ is true.
     - $\exists x\, P(x)$ is also true.
  2. $Q(x)$: "$x^2 = 2$"
     - $\forall x\, Q(x)$ is false.
     - $\exists x\, Q(x)$ is true (for $x = \sqrt{2}$ or $x = -\sqrt{2}$).
  3. $R(x)$: "$x^2 = -1$"
     - $\forall x\, R(x)$ is false.
     - $\exists x\, R(x)$ is false (over the reals, but true over the complex numbers).

# Quantifiers with Two or More Variables

- When a predicate has more than one variable, each variable can have its own quantifier.
- Example:

$$P(x, y): \text{“}x < y\text{”}$$

over the domain of integers.

  - $\forall x \forall y\, P(x, y)$: “For all integers $x$ and $y$, $x < y$”. This statement is false.
  - $\forall x \exists y\, P(x, y)$: “For every integer $x$, there exists an integer $y$ such that $x < y$”. This statement is true (for example, $y = x + 1$).
  - $\exists x \forall y\, P(x, y)$: “There exists an integer $x$ such that $x < y$ for all integers $y$”. This statement is false, because there is no smallest integer.

# Order of Quantifiers

- When more than one quantifier appears in a statement, the order of quantifiers matters.
- Example:

$$P(x, y): \text{“}x + y = 0\text{”}$$

over the domain of integers.

  - $\forall x \exists y\, P(x, y)$: “For every integer $x$, there exists an integer $y$ such that $x + y = 0$”. This statement is true (for example, $y = -x$).
  - $\exists y \forall x\, P(x, y)$: “There exists an integer $y$ such that for every integer $x$, $x + y = 0$”. This statement is false, since no single $y$ works for all $x$.

# Negation of Quantified Statements

- The negation of quantified statements follows specific logical rules:

| Statement | Negation |
|---|---|
| $\neg(\forall x\, P(x))$ | $\exists x \neg P(x)$ |
| $\neg(\exists x\, P(x))$ | $\forall x \neg P(x)$ |

- In words:
  - The negation of "All $x$ have property $P$" is "There exists an $x$ that does not have property $P$".
  - The negation of "There exists an $x$ with property $P$" is "All $x$ do not have property $P$".

# Negation of Quantified Statements: Examples

- $\forall x\, P(x)$: "All swans are white."

  Negation: $\exists x \neg P(x)$: "There exists at least one swan that is not white."
- $\exists x\, Q(x)$: "There exists a student who passed the exam."

  Negation: $\forall x \neg Q(x)$: "All students failed the exam."

# Combining Quantifiers and Logical Connectives

- Quantifiers can be combined with logical connectives such as $\wedge$, $\vee$, and $\rightarrow$.
- Example:

$$\forall x\,(P(x) \wedge Q(x))$$

means that both $P(x)$ and $Q(x)$ are true for every $x$.

- Example:

$$\exists x\,(P(x) \vee Q(x))$$

means that there exists at least one $x$ for which $P(x)$ or $Q(x)$ is true.

- Note that

$$\forall x\,(P(x) \wedge Q(x)) \equiv (\forall x\,P(x)) \wedge (\forall x\,Q(x)),$$
$$\exists x\,(P(x) \vee Q(x)) \equiv (\exists x\,P(x)) \vee (\exists x\,Q(x)).$$

# Combining Quantifiers and Logical Connectives (cont.)

- However, similar equivalences do **not** hold when the connectives are exchanged:

$$\forall x\,(P(x) \vee Q(x)) \not\equiv (\forall x\,P(x)) \vee (\forall x\,Q(x)),$$
$$\exists x\,(P(x) \wedge Q(x)) \not\equiv (\exists x\,P(x)) \wedge (\exists x\,Q(x)).$$

- Example:

  Let the domain be the set of people, and let:

  - $P(x)$: "$x$ is tall."
  - $Q(x)$: "$x$ has dark hair."

- $\forall x\,(P(x) \vee Q(x))$ means: "Everyone is tall or has dark hair."
- $(\forall x\,P(x)) \vee (\forall x\,Q(x))$ means: "Everyone is tall, or everyone has dark hair."

  These are clearly different statements.

# Nested Quantifiers: Example 1

- Let $L(x, y)$: "$x$ loves $y$", with domain = people.
  1. $\forall x \exists y\, L(x, y)$: "Everyone loves someone."
  2. $\exists y \forall x\, L(x, y)$: "There is someone who is loved by everyone."
- The order of quantifiers completely changes the meaning of the statement.

# Nested Quantifiers: Example 2

- Let $T(x, y)$: "$x$ teaches $y$", where $x$ represents a teacher and $y$ a subject.
  1. $\forall x \exists y\, T(x, y)$: "Every teacher teaches at least one subject."
  2. $\exists y \forall x\, T(x, y)$: "There is a subject that every teacher teaches."
- Again, the statements have very different meanings.

# Examples with Quantifiers and Connectives

- Let $D(x)$: "$x$ is a doctor" and $M(x)$: "$x$ is married."
  1. $\forall x\,(D(x) \to M(x))$: "All doctors are married."
  2. $\exists x\,(D(x) \wedge M(x))$: "There exists a doctor who is married."
  3. $\neg\forall x\,(D(x) \to M(x))$: "There exists a doctor who is not married."
  4. $\forall x\,(D(x) \to \neg M(x))$: "All doctors are unmarried."

# Equivalence of Quantified Statements

- Two quantified statements are **logically equivalent** if they have the same truth value for every possible interpretation.
- Examples of equivalences involving quantifiers:

$$\neg(\forall x\, P(x)) \equiv \exists x \neg P(x),$$
$$\neg(\exists x\, P(x)) \equiv \forall x \neg P(x),$$
$$\forall x\, (P(x) \to Q(x)) \equiv \neg\exists x\, (P(x) \wedge \neg Q(x)).$$

# Equivalence of Quantified Statements: Example

- Show that $\forall x\,(P(x) \to Q(x))$ and $\neg\exists x\,(P(x) \wedge \neg Q(x))$ are logically equivalent.

  **Solution.**

$$
\begin{aligned}
\forall x\,(P(x) \to Q(x)) &\equiv \neg\exists x\,\neg(P(x) \to Q(x)) && \text{(Negation of the universal quantifier)}\\
&\equiv \neg\exists x\,\neg(\neg P(x) \vee Q(x)) && \text{(Equivalence of conditionals)}\\
&\equiv \neg\exists x\,(P(x) \wedge \neg Q(x)) && \text{(De Morgan's Laws).}
\end{aligned}
$$

  Hence, they are equivalent.

# De Morgan's Laws for Quantifiers

- De Morgan's laws can also be applied to quantified statements:

$$\neg \forall x\, P(x) \equiv \exists x \neg P(x),$$
$$\neg \exists x\, P(x) \equiv \forall x \neg P(x).$$

- These laws allow us to move negations past quantifiers.
- Example:
  - Original statement: "Not everyone is happy."
  - Logical form: $\neg \forall x\, H(x)$
  - Using De Morgan: $\exists x \neg H(x)$ "There exists someone who is not happy."

# Negating Nested Quantifiers

- Negating statements with multiple quantifiers requires careful application of De Morgan's laws:

$$\neg\forall x\exists y\, P(x,y) \equiv \exists x\forall y\, \neg P(x,y),$$
$$\neg\exists x\forall y\, P(x,y) \equiv \forall x\exists y\, \neg P(x,y).$$

- Example:
  - Statement: "Everyone loves someone." $\forall x\exists y\, L(x,y)$
  - Negation: "There is someone who loves no one." $\exists x\forall y\, \neg L(x,y)$

# Summary of Quantifier Rules

- Key rules for manipulating quantified statements:
  1. $\neg \forall x\, P(x) \equiv \exists x\, \neg P(x)$
  2. $\neg \exists x\, P(x) \equiv \forall x\, \neg P(x)$
  3. $\forall x\, (P(x) \wedge Q(x)) \equiv (\forall x\, P(x)) \wedge (\forall x\, Q(x))$
  4. $\exists x\, (P(x) \vee Q(x)) \equiv (\exists x\, P(x)) \vee (\exists x\, Q(x))$
  5. $\forall x\, (P(x) \vee Q(x)) \not\equiv (\forall x\, P(x)) \vee (\forall x\, Q(x))$
  6. $\exists x\, (P(x) \wedge Q(x)) \not\equiv (\exists x\, P(x)) \wedge (\exists x\, Q(x))$
- Always pay attention to the order of quantifiers and the placement of negations.

# Practice Examples

- Translate the following statements into logical form:
  1. "Some students like mathematics."
  2. "All professors are researchers."
  3. "There is a book that everyone has read."
  4. "Not all animals are mammals."
  5. "Everyone in the class has solved at least one problem."
- Then, practice negating them using De Morgan's laws.

# Logical Equivalences Involving Quantifiers

- Predicates statements are logically equivalent if they always have the same truth value, regardless of the domain or predicate content.
- Common equivalences:

$$\forall x\, P(x) \equiv \neg \exists x \neg P(x),$$
$$\exists x\, P(x) \equiv \neg \forall x \neg P(x),$$
$$\forall x\, (P(x) \wedge Q(x)) \equiv (\forall x\, P(x)) \wedge (\forall x\, Q(x)),$$
$$\exists x\, (P(x) \vee Q(x)) \equiv (\exists x\, P(x)) \vee (\exists x\, Q(x)).$$

- Be careful:

$$\forall x\, (P(x) \vee Q(x)) \not\equiv (\forall x\, P(x)) \vee (\forall x\, Q(x)),$$
$$\exists x\, (P(x) \wedge Q(x)) \not\equiv (\exists x\, P(x)) \wedge (\exists x\, Q(x)).$$

# Conditional Universal Statements

- A universal conditional statement has the form

$$\forall x\,(P(x) \to Q(x))$$

- Examples:

1. "If a real number is greater than 2, then its square is greater than 4"

$$\forall x \in \mathbb{R}\,((x > 2) \to (x^2 > 4))$$

2. "Every byte has eight bits"

$$\forall x\,(x \text{ is a byte} \to x \text{ has eight bits})$$

- Often written equivalently as

$$\forall x \in D\,P(x) \equiv \forall x\,(x \in D \to P(x))$$

# Negation of Conditional Universals

- To negate a universal conditional:

$$\begin{aligned}\neg\forall x\,(P(x) \to Q(x)) &\equiv \exists x \neg(P(x) \to Q(x)) \\ &\equiv \exists x \neg(\neg P(x) \vee Q(x)) \\ &\equiv \exists x\,(P(x) \wedge \neg Q(x))\end{aligned}$$

- Examples:
  - "Every blonde person has blue eyes" Negation: "There exists a blonde person who does not have blue eyes"
  - "If a program is written in C, it has a bug" Negation: "There exists a program written in C that does not have a bug"

# Vacuous Truth in Universal Statements

- If the premise $P(x)$ of an implication is false, then

$$P(x) \rightarrow Q(x)$$

is always true, regardless of $Q(x)$.

- Therefore, if $P(x)$ is false for every $x$ in the domain $D$, then

$$\forall x \in D\,(P(x) \rightarrow Q(x))$$

is vacuously true.

# Example of Vacuous Truth

- Consider five blue balls, five white balls, and a plate.
  - **Scenario 1:** three blue balls and one white ball are placed on the plate.

    Statement: "All balls on the plate are blue"

    Negation: "There exists at least one ball on the plate that is not blue"

    Thus, the statement is false.

# Example of Vacuous Truth

- **Scenario 2:** no balls are placed on the plate.

  Statement: "All balls on the plate are blue"

  Negation: "There exists at least one ball on the plate that is not blue"

  Thus, the statement is true (vacuously).

- Another perspective:

For every ball, if it is on the plate, then it is blue.

The hypothesis is false for every ball, making the implication true.

# Inference Rules: Introduction

- In many situations, it is necessary to deduce **conclusions** from **premises**:
  1. in mathematics: to establish absolute truths (theorems),
  2. in computer science: to verify that the properties of a system are satisfied given its specification,
  3. in politics/philosophy: to demonstrate that certain ideas are well-founded.
- The process of deriving conclusions from premises is called an **argument**.

  An argument is **valid** if and only if it is impossible for all its premises to be true and its conclusion to be false.
- Here we will study **inference rules** that allow us to derive valid arguments.
- We will also study common reasoning errors, called **fallacies**.

# Inference Rules

# Valid Arguments in Propositional Logic

- An **argument** is a sequence of propositions.

  All statements except the last are called **premises**.

  The final proposition is called the **conclusion**.

  A **valid argument** is one in which the truth of its premises implies the truth of its conclusion.

- An **argument form** in propositional logic is a sequence of propositions involving propositional variables.

  A **valid argument form** is one in which, regardless of the values assigned to each propositional variable in the premises, the conclusion is true whenever all the premises are true.

# Valid Arguments in Propositional Logic

- Consider the following argument involving propositions:

  1. "If you are enrolled in Discrete Mathematics, then you have access to the course webpage."
  2. "You are enrolled in Discrete Mathematics."

     Therefore,

  3. "You have access to the course webpage."

  Is the argument above valid?

  In other words, is it true that the conclusion (3) is true whenever the premises (1) and (2) are both true?

# Valid Arguments in Propositional Logic

- **Solution.** Let's analyse the argument form.

  Let

  $p$ be the proposition "You are enrolled in Discrete Mathematics", and

  $q$ be the proposition "You have access to the course webpage".

  The previous argument has the form

$$\frac{p \rightarrow q \quad p}{q}$$

# Valid Arguments in Propositional Logic

- Considering $p$ and $q$ as propositional variables, we can use a truth table to verify that whenever the premises are true, the conclusion is also true.

| | | Premises | | Conclusion |
|---|---|---|---|---|
| $\mathbf{p}$ | $\mathbf{q}$ | $\mathbf{p \to q}$ | $\mathbf{p}$ | $\mathbf{q}$ |
| $T$ | $T$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $T$ | $F$ |
| $F$ | $T$ | $T$ | $F$ | $T$ |
| $F$ | $F$ | $T$ | $F$ | $F$ |

The table shows that whenever $p \to q$ and $p$ are both true, $q$ must also be true. Thus, the argument form

$$\frac{p \to q \quad p}{q},$$

is valid since whenever the premises are true, the conclusion is true.

# Valid Arguments in Propositional Logic

- A valid argument can be interpreted as a rule of truth preservation:
  1. If all premises are true, you can apply the argument to reach a true conclusion.
  2. If the premises are not all true, applying the argument may lead to either a true or a false conclusion.

# Valid Arguments in Propositional Logic

- Consider the following argument involving propositions:
  1. "If you are watching this class, you are the most famous person in the UK."
  2. "You are watching this class."

     Therefore,
  3. "You are the most famous person in the UK."

  Is the argument above valid? Is its conclusion true?

# Valid Arguments in Propositional Logic

- **Solution.** Let's analyse the argument form. Let

  $p$ be the proposition "You are watching this class", and

  $q$ be the proposition "You are the most famous person in the UK".

  The argument is valid because it has the form

$$\frac{p \to q \quad p}{q},$$

which we have already verified to be valid: whenever the premises are true, the conclusion will also be true.

However, in the argument above, premise (1) is false, so the conclusion is not guaranteed to be true. In fact, the conclusion is probably false!

# Valid Arguments in Propositional Logic

- An **invalid argument form** is one in which the premises can all be true and yet the conclusion can be false.
- Show that the following argument form is invalid.

$$\frac{p \to q \quad q}{p}$$

**Solution.**

The third line of the truth table shows that the premises $p \to q$ and $q$ can both be true, while the conclusion $p$ is false. Thus, the argument form is invalid.

| | | Premises | | Conclusion |
|---|---|---|---|---|
| **p** | **q** | **p → q** | **q** | **p** |
| *T* | *T* | *T* | *T* | *T* |
| *T* | *F* | *F* | *F* | *T* |
| *F* | *T* | *T* | *T* | *F* |
| *F* | *F* | *T* | *F* | *F* |

For example, from the premises "If John wins the lottery, he becomes rich" and "John is rich", we cannot conclude that necessarily "John won the lottery".

# Valid Arguments in Propositional Logic: Validity vs. Truth

- Validity is a property of the argument form.
- Truth is a property of the argument's conclusion.

"If Australia is a rich country, then it is in the northern hemisphere." "Australia is a rich country."
---
"Australia is in the northern hemisphere."

is a valid argument, but the first premise is false, and the conclusion is also false.

"If London is a metropolis, then it has tall buildings." "London has tall buildings."
---
"London is a metropolis."

is an invalid argument, but the conclusion is true.

# Inference Rules for Propositional Logic

- To show that an argument is valid, we can always use a truth table to check whether, whenever its premises are true, the conclusion is also true.
- However, if the argument has many propositional variables, constructing a truth table can be inconvenient.

  For example, to verify the validity of an argument involving 10 propositional variables, one would need a table with $2^{10} = 1,024$ rows.
- To overcome this problem, we verify the validity of relatively simple arguments, called **inference rules**, and use these rules to build more complex arguments consistently.

# Inference Rules for Propositional Logic

- The argument established in the previous slides, for example, consists of the inference rule called **modus ponens** (from Latin, mode of affirmation):

$$\frac{p \quad p \rightarrow q}{q}$$

The rule of modus ponens tells us that:

1. if a conditional statement $p \rightarrow q$ is true,

   and

2. the hypothesis $p$ of the conditional is true,

   then

3. the conclusion $q$ of the conditional is necessarily true.

# Inference Rules for Propositional Logic

- Suppose we know that
  1. "If it is sunny today, I will go to the club",

     and that
  2. "It is sunny today",

     then, by modus ponens, we can conclude that
  3. "I will go to the club."

# Inference rules for the propositional logic

- Inference rules for the propositional logic:

| Name | Inference rule |
|---|---|
| **Modus ponens** | $\frac{p \rightarrow q \quad p}{q}$ |
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| **Modus tollens** | $\frac{p \rightarrow q \quad \neg q}{\neg p}$ |
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| **Modus ponens** | $\frac{p \to q \quad p}{q}$ |
| **Modus tollens** | $\frac{p \to q \quad \neg q}{\neg p}$ |
| **Hypothetical syllogism** | $\frac{p \to q \quad q \to r}{p \to r}$ |
| | |

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| **Disjunctive syllogism** | $\frac{p \vee q \quad \neg p}{q}$ |

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| Name | Inference rule |
|---|---|
| **Addition** | $\frac{p}{p \vee q}$ |
| | |
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| | |

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| Name | Inference rule |
|---|---|
| **Addition** | $\frac{p}{p \vee q}$ |
| **Simplification** | $\frac{p \wedge q}{p}$ |
| | |
| | |

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| **Hypothetical syllogism** | $\frac{p \to q \quad q \to r}{p \to r}$ |
| **Disjunctive syllogism** | $\frac{p \vee q \quad \neg p}{q}$ |

| Name | Inference rule |
|---|---|
| **Addition** | $\frac{p}{p \vee q}$ |
| **Simplification** | $\frac{p \wedge q}{p}$ |
| **Conjunction** | $\frac{p \quad q}{p \wedge q}$ |
| | |

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| **Disjunctive syllogism** | $\frac{p \vee q \quad \neg p}{q}$ |

| Name | Inference rule |
|---|---|
| **Addition** | $\frac{p}{p \vee q}$ |
| **Simplification** | $\frac{p \wedge q}{p}$ |
| **Conjunction** | $\frac{p \quad q}{p \wedge q}$ |
| **Resolution** | $\frac{p \vee q \quad \neg p \vee r}{q \vee r}$ |

# Inference Rules for Quantified Propositions

- Show that the premises
  1. "All students enrolled in Discrete Mathematics are dedicated students"

     and
  2. "Alice is enrolled in Discrete Mathematics"

     imply the conclusion
  3. "Alice is a dedicated student"

  **Solution.** Let's define the following predicates, with the domain being the set of all students:

  - $M(x)$ : "$x$ is enrolled in Discrete Mathematics."
  - $D(x)$ : "$x$ is a dedicated student."

# Inference Rules for Quantified Propositions

- The premises of the argument are, therefore:

  1. $\forall x : (M(x) \rightarrow D(x))$
  2. $M(\texttt{Alice})$

  The conclusion of the argument is:

  (c) $D(\texttt{Alice})$

  The derivation of the conclusion from the premises can be done as follows:

  (1)
  $$\frac{\forall x : (M(x) \rightarrow D(x))}{M(\texttt{Alice}) \rightarrow D(\texttt{Alice})}$$
  *Premise (a)*
  *Universal instantiation*

  (2)
  $$\frac{\begin{array}{l} M(\texttt{Alice}) \\ M(\texttt{Alice}) \rightarrow D(\texttt{Alice}) \end{array}}{D(\texttt{Alice})}$$
  *Premise (b)*
  *Conclusion of (1)*
  *Modus ponens*

# Inference Rules for Quantified Propositions

- Show that the premises
  1. "A Discrete Mathematics student did not read the textbook", and
  2. "All Discrete Mathematics students did well on the exam"

  imply the conclusion
  3. "Someone who did well on the exam did not read the textbook"

  **Solution.** Let's define the following predicates, with the domain being the set of all people:
  - $D(x)$ : "$x$ is a Discrete Mathematics student."
  - $L(x)$ : "$x$ read the textbook."
  - $P(x)$ : "$x$ did well on the exam."

# Inference Rules for Quantified Propositions

- The premises of the argument are, therefore:

  1. $\exists x : (D(x) \wedge \neg L(x))$
  2. $\forall x : (D(x) \rightarrow P(x))$

  The conclusion of the argument is:

  (c) $\exists x : (P(x) \wedge \neg L(x))$

  The derivation of the conclusion from the premises can be done as follows:

  (1) $\dfrac{\exists x : (D(x) \wedge \neg L(x))}{D(c) \wedge \neg L(c)}$ *Premise (a)* / *Existential instantiation*

  (2) $\dfrac{D(c) \wedge \neg L(c)}{D(c)}$ *Conclusion of (1)* / *Conjunctive simplification*

  (3) $\dfrac{\forall x : (D(x) \rightarrow P(x))}{D(c) \rightarrow P(c)}$ *Premise (b)* / *Universal instantiation*

# Inference Rules for Quantified Propositions

- (4)
$$\begin{array}{ll} D(c) & \textit{Conclusion of (2)} \\ D(c) \rightarrow P(c) & \textit{Conclusion of (3)} \\ \hline P(c) & \textit{Modus ponens} \end{array}$$

(5)
$$\begin{array}{ll} D(c) \wedge \neg L(c) & \textit{Conclusion of (1)} \\ \hline \neg L(c) & \textit{Conjunctive simplification} \end{array}$$

(6)
$$\begin{array}{ll} P(c) & \textit{Conclusion of (4)} \\ \neg L(c) & \textit{Conclusion of (5)} \\ \hline P(c) \wedge \neg L(c) & \textit{Conjunctive addition} \end{array}$$

(7)
$$\begin{array}{ll} P(c) \wedge \neg L(c) & \textit{Conclusion of (6)} \\ \hline \exists x : (P(x) \wedge \neg L(x)) & \textit{Existential generalization} \end{array}$$

# Combining Inference Rules for Propositions and Quantified Predicates

- We have developed inference rules for propositions and for quantified propositions.
- Since modus ponens and universal instantiation are frequently used together, we can use the following rule known as **universal modus ponens**:

$$\frac{\forall x : (P(x) \rightarrow Q(x)) \quad P(c), \text{ where } c \text{ is a particular element of the domain}}{Q(c)}$$

- Assume the following premise is true: "For every positive integer $n$, if $n > 4$, then $n^2 < 2^n$".

  We know that the proposition "$100 > 4$" is true.

  By universal modus ponens, we can conclude that "$100^2 < 2^{100}$".

# Addendum: Fallacies

# Fallacies: Introduction

- A **fallacy** is an invalid argument that appears valid.
- The term "fallacy" derives from the Latin verb "fallere", which means "to deceive".
- Arguments intended to persuade may appear convincing to a large portion of the audience and still be invalid, thus constituting fallacies.
- Recognising fallacies is sometimes difficult.

  Fallacious arguments may have emotional, intimate, or psychological validity, but not logical validity.

# Fallacious Arguments

- The mere fact that someone commits a fallacy does not invalidate all of their argumentation.

  For example, one cannot conclude that:

  "I listened to the debate between professors Alice and Bob, but Alice committed a fallacy and therefore all her conclusions must be wrong."

- A fallacy immediately invalidates the argument in which it occurs.
- However, a fallacy does not necessarily invalidate the conclusion of the argument.

  Only the fallacious argument is discarded from the reasoning.

  There may be other arguments that succeed in proving the truth of the same conclusion.

# Formal and Informal Fallacies

- Fallacies can be classified as formal and informal.
- A **formal fallacy** is an invalid argument because its form is invalid.

  Examples:

  1. **Affirming the consequent, or converse error**:

     "All English nationals are British. Maria is British. Therefore, Maris is from England."

  2. **Denying the antecedent, or inverse error**:

     "All English nationals are British. Maria is not from England. Therefore, Maria is not British."

- An **informal fallacy** is an invalid argument since, although its form is valid, its premises are not all true.

# Types of Fallacies

- There are several very common types of fallacies.

  Knowing fallacies can help you:

  - argue your ideas better;
  - avoid being deceived by invalid arguments; and
  - not embarrass your logic professor online (like on Facebook)!

- **Important note:**

  In the following examples it does not matter whether the professor agrees or disagrees with the conclusions of the arguments.

  It does not even matter if the conclusions are true or false.

  Only the fact that the reasoning used does not guarantee the validity of the conclusions matters, and therefore, the arguments are fallacious.

# Attacking the interlocutor (Ad hominem)

- General form: because the person arguing for the proposition is not trustworthy, the proposition must be false.
  1. "Look who's saying that drinking is harmful: the drunk in the room!"
  2. "Look at the stupid face of this person; you don't even need to listen to what they say to know it's all wrong!"
  3. "To see that candidate Montenegro's economic proposal is unfeasible, just look at his history of mismanagement."
- How to avoid it: The truth of a proposition does not depend on who states it.

  Anyone can argue for or against a proposition, as long as the argument is consistent.

  Analyse the argument, not the arguer.

# Appeal to the majority/popularity (Ad populum)

- General form: since the majority believes a proposition is true, it must necessarily be true.

  1. "The majority of British are against (privatizations/ lowering the age of criminal responsibility/abortion/euthanasia), therefore the practice is wrong."
  2. "Of course (Christianity/Judaism/Spiritism/Atheism/Pastafarianism) is true.

     Do you really think that the (10 thousand/10 million/1 billion) people who believe in it could all be wrong?"

- How to avoid it: no matter how popular a proposition is, only the evidence supporting it matters.

  It was once almost universally believed that all numbers were rational (poor Greeks, so disappointed!), that the Sun revolved around the Earth, that gods inhabited Mount Olympus ...

  Truth is not decided by vote.

# Begging the Question (Petitio principii)

- General form: one assumes as a premise exactly what one wants to prove.

  1. "Abortion is wrong because it is a vile act."
  2. "Abortion is not wrong because it is a human right."

  (In both cases above, one proves that abortion is good or bad by already assuming that it is good or bad.)

- How to avoid it: to prove your point, start from premises that the interlocutor admits are true, and only from there construct your argument.

  If these premises have not yet been found, you need to go back to even more basic statements on which both sides agree in order to construct the argument from them.

  Reasoning must start from hypotheses accepted by both you and your interlocutor.

# Appeal to Ignorance

- General form: because you have no better explanation than mine, my explanation must be correct.

  1. "You don't suggest any suspect besides Alice, so you are forced to conclude she is guilty."
  2. "You say thunder is not produced by Zeus, so provide a better explanation.

     A-ha, since you have no better explanation, it must be Zeus creating the thunder!"

- How to avoid it: The burden of proof lies with the proposer, not the challenger.

  It is logically more correct to recognise that we have no explanation for a phenomenon than to accept an explanation without evidence.

# Other Fallacies

- There is a large number of fallacies:
  - false dilemma;
  - appeal to antiquity/authority/emotion;
  - straw man;
  - reversal of cause and effect;
  - non-sequitur;
  - "after this, therefore because of this";
  - ...
- Fallacies appear in situations of various levels of seriousness: in politics, in medicine, in statements from fan-voters, and in opinions of friends on Facebook.

  We are all subject to committing or being deceived by fallacies!

  Knowing them can help you argue better and not be misled by invalid arguments.